Rodrigo da Motta Cabral de Carvalho

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Rodrigo-Motta Personal Page

Education

MS - Neuroscience and Cognition, **Federal University of ABC** — Grande São Paulo, SP, Brazil, Sep 2023 – July 2025

- **Grade**: A/A;
- **Thesis**: A Graph Neural Network Approach to Investigate Brain Critical States.

MS Visiting Research Student - Biomedical Engineering & Imaging Sciences, King's College London — Greater London, UK, Jul 2024 – Mar 2025

- Awarded a prestigious scholarship to work at the MeTrICS Lab (Machine Learning for Translational Imaging and Cortical Surfaces).

BS - Physics, University of São Paulo — São Paulo, SP, Brazil, Jan 2019 – Dec 2022

- Best university in South America.
- **Grade**: 8.6/10 (Top 15%)
- **Extracurricular:** Community-Building in Data Science at Hackerspace.IFUSP

Experience

Research Scientist, **CloudWalk Inc.** — São Paulo, SP, BR, April 2024 – Ongoing

CloudWalk is an AI Fintech unicorn where I:

- Work on transformer-based representation learning of clients for in-house financial foundation models, also developing post-training for downstream tasks and steering for data drift.
- Led a team to develop a self-healing multi-agentic systems for auto research and for a self-driven vending machine. The vending machine is operating with almost no human in the loop for 8 months now.
- Develop post-training pipelines for robotics foundation models, with a focus on humanoids. Dataset and model checkpoints: ClouWalk Robotics Lab.
- Led a project that fine-tune LLMs with data from altered states of consciousness to explore shifts in internal representations. Using information theory and embedding dynamics to draw connections between machine and human cognition. Developed new psychometric evaluation methods for LLMs and AI agents, evaluating more specific measures than classical benchmarks.

Research Intern, **King's College London** — Greater London, UK, Jul 2024 – March 2025

Advanced AI applications in psychiatry at MeTrICS Lab by applying Deep Learning models (e.g., Spatial-Temporal Graph Neural Networks) on GPU clusters to classify psychiatric disorders using large-scale fMRI data from a Brazilian cohort. Achieved high diagnostic accuracy and identified key brain network contributions relating to psychiatric scores.

- **Advisor:** Dr. Emma Claire Robinson – King's College London (KCL)
- **Co-Advisor:** Dr. Walter Pinaya - Hologen AI & KCL

Graduate Research Scholar, **FAPESP** — São Paulo, SP, BR, Sep 2023 – March 2025

Applied LLMs for semantic analysis in psychology, focused on semantic trajectory over neurodegenerative diseases and time perception. Also developed Graph Neural Networks with complex systems simulations to model brain dynamics under acute and subacute effects of ayahuasca. Managed the project in an international team in a open-source project.

- **Advisor:** Prof. Dr. João Ricardo Sato - Federal University of ABC (UFABC)
- **Co-Advisor:** Dr. Walter Pinaya - Hologen AI & KCL

Data Scientist, **Artefact** — São Paulo, SP, BR, Jul 2022 – Sept 2023

Enabled business intelligence for the North American tire market by developing and deploying executive-level forecasting models for all tire SKUs and estimated variables impact, such as inflation. Managed the end-to-end data pipeline—from accessing raw data to final model deployment. Additionally, led community-building initiatives by designing a Data Science meetup and the global journal clubs.

Undergraduate Researcher, **Brazilian Institute of Neuroscience and Neurotechnology (BRAINN)** — Campinas, SP, BR, Jul 2021 – Jul 2022

Employed complex system models to better understand brain rest state hemodynamics with functional near-infrared spectroscopy. Developed open-source tools and methods.

- **Advisor:** Prof. Dr. Rickson Coelho Mesquita - UNICAMP & University of Birmingham

Publications

Toro-Hernandez, F., Filho J., **Cabral-Carvalho, R.** (2026). Characterizing Human Semantic Navigation in Concept Production as Trajectories in Embedding Space. The Fourteenth International Conference on Learning Representations (ICLR) 2026, https://openreview.net/forum?id=QQVmIR97sf — 2026

Cabral-Carvalho, R., Pinaya, W. H. L., Sato, J. R. (2025). A Graph Neural Network Approach to Investigate Brain Critical States Over Neurodevelopment. *Network Neuroscience* (Cambridge, Mass.), 9(2), 761–776. https://doi.org/10.1162/netn_a_00451 — 2025

Falconi-Souto, A., **Cabral-Carvalho, R.M.**, Fujita, A. et al. (2025). Inferences on the Watts-Strogatz Model: A Study on Brain Functional Connectivity. *Neuroinform* 23, 57. https://doi.org/10.1007/s12021-025-09756-z — 2025

Paris-Colombo, G., **Cabral-Carvalho, R.M.**, Toro-Hernández, F.D. (2026). Comparing Semantic Navigation in Humans and Large Language Models using Natural Language Processing. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 48. arXiv:2607.12195 — 2026

Pre-prints

Cabral-Carvalho, R. M., Palhano-Fontes, F., Araujo, D. B., Sato, J. R. (2025). Ayahuasca shifts brain dynamics toward higher entropy: Persistent elevation of Ising temperature correlates with acute subjective effects. *bioRxiv*. https://doi.org/10.1101/2025.04.25.650509v1 (Under review at *Network Neuroscience*). — 2025

Lelis, B., **Cabral-Carvalho, R. M.** (2026). RCWT: Measuring Task-Budget Displacement from Coordination Content in LLM Calls. *arXiv preprint arXiv:2607.12216*. https://doi.org/10.48550/arXiv.2607.12216 — 2026

Bonifácio, T. A. S., **Cabral-Carvalho, R.**, Cravo, A. (2024). Self-report measures of subjective time: An overview of existing measures and their semantic similarities. *PsyArXiv*. https://doi.org/10.31234/osf.io/sjwm2 — 2024

Conferences, Talks and funding

Scholarships & Funding

- Undergraduate Researcher (FAPESP Scholarship),
- Graduate Researcher (FAPESP Scholarship),
- KCL Research Internship Abroad (BEPE Scholarship): 20k USD.
- CloudWalk AI Research Grant: 6k USD

Conferences:

- Publication and Poster at **ICLR 2026** (Rio de Janeiro, Brazil)
- Poster at **Stanford Graph Learning Workshop**: AgentMarket: Generative Agent in Dynamic Market Networks (Palo Alto, USA)
- Talk at **Brazilian Society of Neuroscience and Behaviour 2025**: Quantifying Subjective Experiences with LLMs (São Paulo, Brazil)
- Talk at **Brain Modes 2025** (Toronto, Canada)
- Poster at **Brain Modes 2024** (Bilbao, Spain)
- Poster at **Organization for Human Brain Mapping (OHBM) 2024** (Seoul, South Korea)
- Oral presentation at **10th Brazilian Institute for Neuroscience and Neurotechnology (BRAINN) Congress 2024** (UNICAMP, Brazil)

Awards

- Best Oral Presentation, 10th BRAINN Congress 2024 (UNICAMP, Brazil)

Invited Talks (on site):

- **Yonsei University, South Korea** (Prof. Byung-Hoon Kim, NAIPL)
- **Imperial College London** (Dr. Pedro Mediano)
- **University of Oxford** (Prof. Rui Costa & NeuroAI group)
- **University of Zurich** (Prof. Susanne Wegener & Prof. Nicolas Langer)

Extra Activities

Content Creator (blog posts) — Jan 2023 - Ongoing

- As a writer, I am inspired to communicate and spark curiosity about Neuroscience, Cognition, Physics, Data Science and AI. Content.

Data Science Coordinator Hackerspace.IF-USP — 2020 - Ongoing

Organize free seminars and tech courses at USP, including a week-long Deep Learning intensive featuring 5 expert speakers for 60 participants. [Course Content]

Languages

English: IELTS 7.5 (Academic) **Portuguese:** Native **Spanish:** Basic (No proficiency)